



CUT LABOR COSTS & PRODUCT RECALLS USING VISION SYSTEMS

LABOR-COST REDUCTION

PRODUCT-RECALL PREVENTION



**High-Speed,
Multi-Camera
Machine Vision System**
CV-X Series

**WORLD STANDARD
CV-X Series**



WHEN and WHERE the Vision System can be used?



Before

REDUCING LABOR COSTS

VISUAL INSPECTIONS



- Disadvantages of visual inspections
- Different inspection results depending on inspectors
- Products may be shipped without inspections



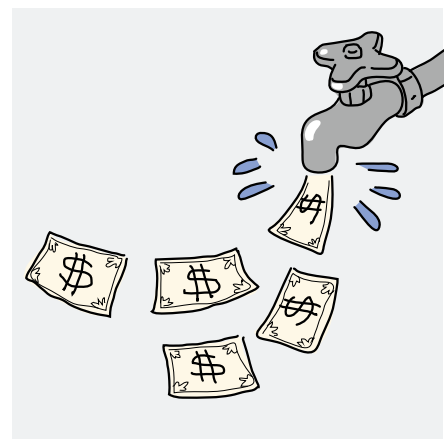
- Increases in labor costs according to expanded production
- Ineffective manual palletizing

REDUCING CUSTOMER COMPLAINTS

END USER WARRANTY CLAIMS...



- Trust issue due to outflow of defective product (lose customers' trust)
- The possibility of direct loss, such as business suspension
- After collecting lots in question, enormous man-hours for inspections including 100% inspections will be needed



- Costs for preventive actions
- Costs are needed for prompt actions to investigate the cause for outflow of defective products and to prevent recurrence
- Costs for collecting and inspecting shipped defective products



**Awesome
Improvement !**

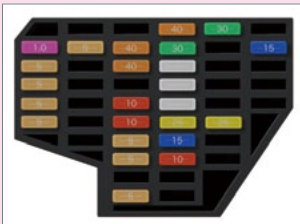
After

REASONS TO INTRODUCE VISION SYSTEMS TO REDUCE INSPECTORS

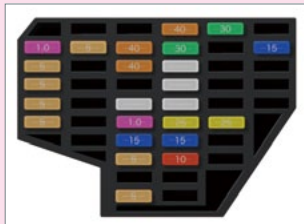
False assembly inspection for fuse boxes



OK image



NG image



Detection of grease application amount



OK image



NG image

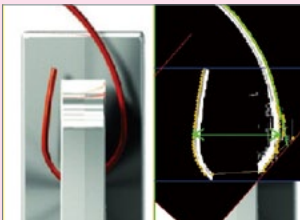


REASONS TO INTRODUCE VISION SYSTEMS TO DECREASE COMPLAINTS FROM END USERS

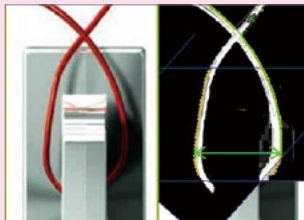
Winding defect inspection for motor wire



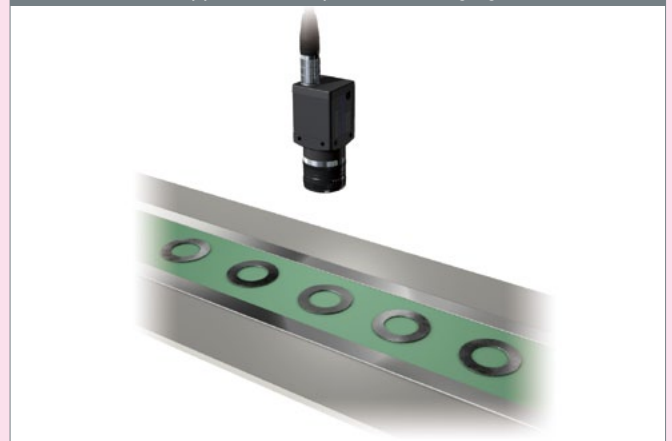
OK image



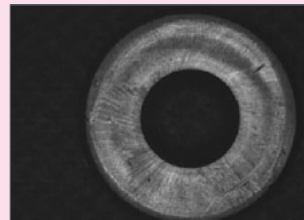
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Appearance inspection for forging



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NG image



Visual inspections are automated to perform the following applications

Black point/
stain inspection

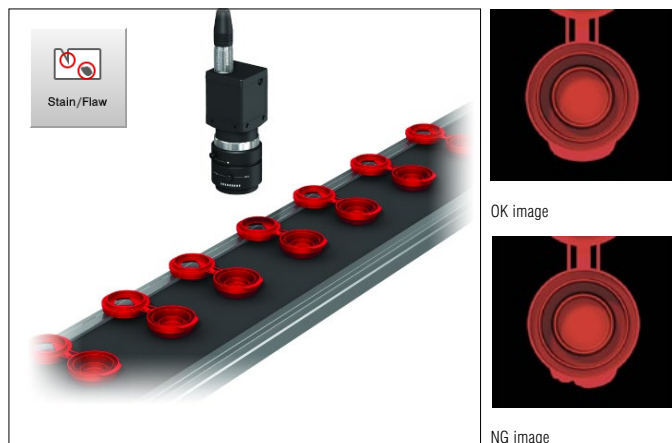
Shape abnormality
inspection

Inspecting scars
on metal surface

Reading of barcodes/
characters

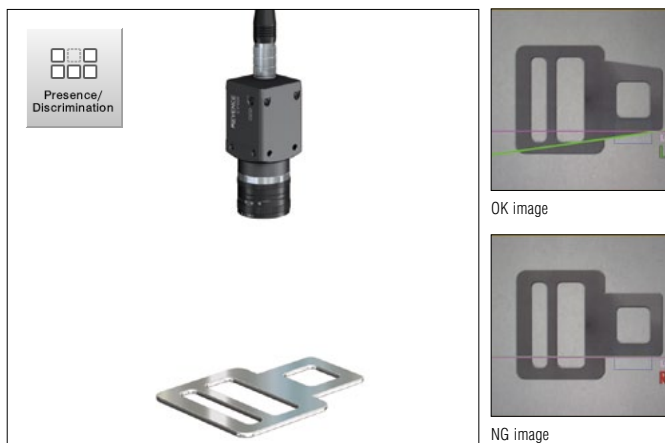
STAIN/FLOW

Appearance inspection for a cap



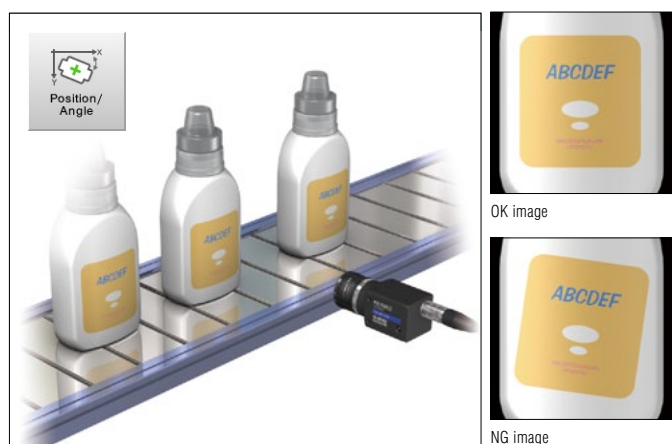
PRESENCE/DISCRIMINATION

Type differentiation for seat belt buckles



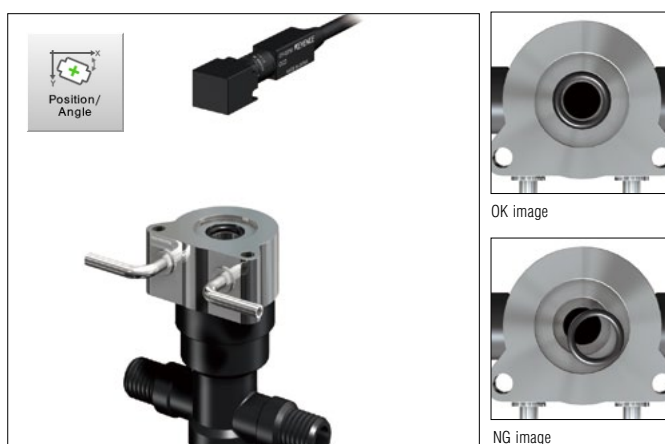
POSITION/ANGLE

Positioning inspection for labels



POSITION/ANGLE

Position detection for O-ring



KEYENCE

Please visit: www.keyence.com



SAFETY INFORMATION

Please read the instruction manual carefully in order to safely operate any KEYENCE product.

GLOBAL NETWORK

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